

CliniScan:Lung-Abnormality Detection on Chest X-rays using AI

1. Dataset Overview

The **VinBigData Chest X-ray Abnormalities Detection** dataset is a large-scale medical imaging dataset available on **Kaggle** for object detection tasks in chest radiographs.

Key Details:

- Contains **15,000+ chest X-ray images** in DICOM format
- Includes **14 disease/abnormality classes** such as:
 - Atelectasis
 - Cardiomegaly
 - Pleural Effusion
 - Pneumothorax
 - Lung Opacity
 - (and other thoracic abnormalities)
- Also includes a **“No Finding”** class
- Annotations are provided as **bounding boxes in CSV format**
- Images are labeled by **multiple expert radiologists**

This dataset is designed for training and evaluating **object detection models** in medical imaging.

2. Data Preprocessing Summary

To prepare the dataset for YOLO-based object detection, the following preprocessing steps were performed:

- Installed required libraries such as **pydicom, OpenCV, NumPy, Pandas, and tqdm**
- Converted the first **200 DICOM images** into PNG format for manageable processing
- Loaded the annotation CSV file
- Created a **class ID mapping dictionary**
- Converted bounding box coordinates into **YOLO format (normalized values)**
- Skipped **“No Finding”** cases with missing (NaN) bounding boxes
- Saved annotations as .txt label files

- Verified correctness by **visualizing bounding boxes** on sample images

Outcome

The dataset was successfully converted into **YOLO-compatible format**, including:

- Processed images (.png)
- Corresponding label files (.txt)
- Verified bounding box alignment